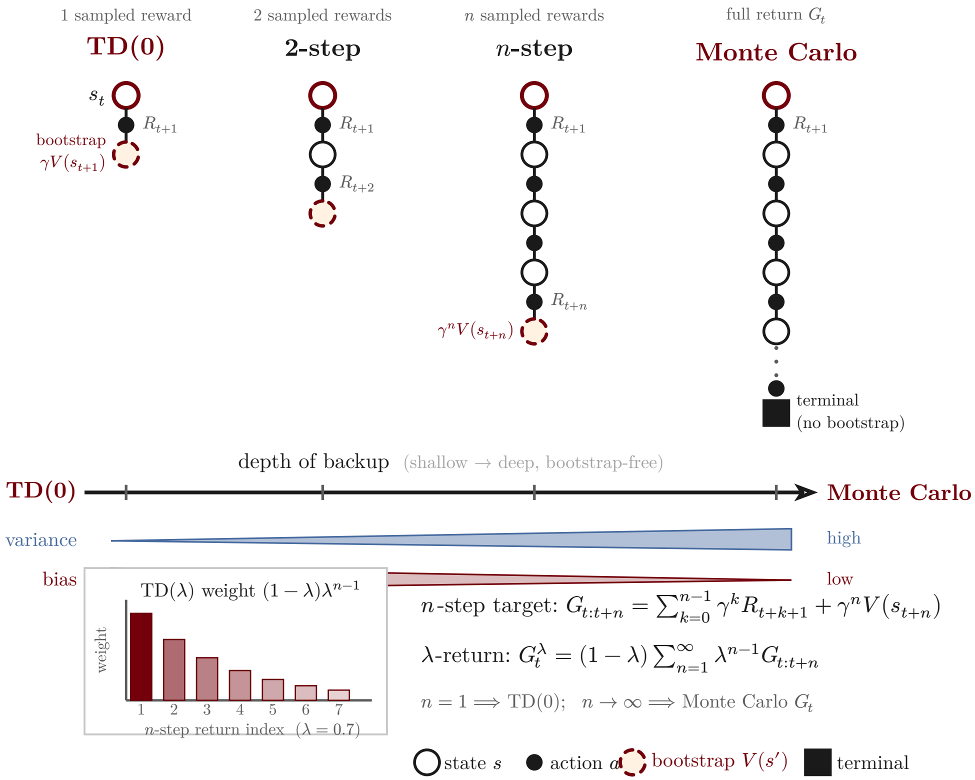

1 sampled reward
TD(0)
2 sampled rewards
2-step
n sampled rewards
n-step
full return G_t
Monte Carlo
s_t
R_{t+1}
bootstrap
$\gamma V(s_{t+1})$
R_{t+2}
R_{t+n}
$\gamma^n V(s_{t+n})$
terminal
(no bootstrap)
depth of backup
(shallow → deep, bootstrap-free)
TD(0)
Monte Carlo
variance
high
bias
low
TD(λ) weight $(1-\lambda)\lambda^{n-1}$
weight
1
2
3
4
5
6
7
n-step return index $(\lambda = 0.7)$
n-step target: $G_{t:t+n} = \sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n V(s_{t+n})$
λ-return: $G_t^\lambda = (1-\lambda)\sum_{n=1}^{\infty} \lambda^{n-1} G_{t:t+n}$
$n = 1 \implies$ TD(0); $n \to \infty \implies$ Monte Carlo G_t
state s
action a
bootstrap $V(s')$
terminal